25901111

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTG202 – TISSUE CULTURE TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Which equipment is used in a plant tissue culture laboratory for ensuring sterile conditions during inoculation? [K] / [CO1]
   Laminar Airflow Chamber- 2 marks.
2. What are the common challenges faced during meristem culture? [K] / [CO3]
   Any two points-2marks.
3. Write any two steps in vaccine production. [K] / [CO4]
   Culturing of micro-organisms, purification, harvesting-Any two-2 marks.
4. What is the role of Auxins in PTC? [U] / [CO2]
   Callus induction, root induction, elongation-any two – 2 marks
5. What are feeder layers? [K] / [CO3]
   Definition- 2 marks
6. List any two commercial applications of tissue culture techniques. [K] / [CO3]
   Any two applications-2 marks
7. Name the process of formation of organs from callus. [K] / [CO3]
   Indirect organogenesis-2 marks
8. Which serum protein helps in cell attachment in animal tissue culture? [U] / [CO2]
   Albumin, Alpha globulin- any two serum proteins-2 marks
9. Define "Sub-culturing.". [K] / [CO4]
   Definition- 2 marks
10. Which equipment is used to observe cultured cells in animal cell culture lab? [K] / [CO1]
    Phase Contrast microscope/Microscope- 2 marks
11. Why is a CO2 incubator used for some cultures? [U] / [CO1]
    Any two functions of CO2 incubator- 2 marks

12. Define haploid production in plants. [K] / [CO3]

    Definition- 2 marks

[2x10 = 20]

Part B-Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Explain the concept of anchorage dependence in cultured cells.. [U] / [CO4]

    Cell culture- introduction- 1 mark, Anchorage dependent cells- 2 marks, concept- explain- 2 marks

14. What are the limitations of somaclonal variations? [K] / [CO3]

    Somaclonal variation-definition- 2 marks, causes- 2 marks, limitations- 2 marks

15. Comment on the importance of an incubation room in a plant tissue culture laboratory. [U] / [CO1]

    Plant tissue culture laboratory- introduction- 1 mark, basic requirements- 2marks, importance of incubation room- 2 marks

16. Why is moist heat more effective than dry heat? [K] / [CO2]

    Sterilization- introduction- 1 mark, moist heat- definition- 1 mark, dry heat- definition- 1 mark, importance of moist heat over dry heat- 1mark.

17. Explain protoplast fusion. [K] / [CO3]

    Any 4 points- 4 marks.

18. Describe how proper sterilization contributes to reproducible animal cell culture experiments.[U]/[CO2]

    Sterilization- introduction- 1 mark, any 3 points- 3marks

19. Write short essay on cytodifferentiation. [K] / [CO3]

    Cytodifferentiation.- definition- 1 mark, changes- steps- 2 marks, importance-1 mark

[4x5 = 20]

Part C-Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. Discuss the role of organogenesis in plant micro propagation and its benefits for agriculture.[U] / [CO3]

    .Organogenesis- definition- 1 mark, steps- 6 marks, applications-4 marks.

21. Comment on the historical development of plant biotechnology from selective breeding to genetic engineering. [U] / [CO1]

    History of plant tissue culture-Selective breeding- 3 marks, Plant tissue culture and techniques- 4 marks, transgenic plants- 3 marks [10x1 = 10]